

Context-Dependent Relational Reorganization of Heart Rate Dynamics Precedes Spontaneous Ventricular Fibrillation: Systemic Tau and Ordinal RECD Evidence from the Sudden Cardiac Death Holter Database

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Target journals (in order): Chaos, Frontiers in Network Physiology, PLOS Computational Biology, Physical Review E (complex systems / nonlinear dynamics section)

Abstract (draft)

Sudden cardiac death from ventricular fibrillation (VF) remains difficult to anticipate from surface ECG. Classic early-warning signals (EWS) such as rising variance or lag-1 autocorrelation often fail or reverse in real cardiac data because they assume a simple critical slowing down. We introduce a relational, multivariate framework (CCTP) combining (i) Systemic Tau (τ_s), a normalized measure of cross-variable coupling change, with (ii) ordinal recurrence quantification (RECD) that decomposes the signal into three hierarchical levels of symbolic structure (Φ_1, Φ_2, Φ_3) and tracks the excess contribution of the highest level (excess3).

Applied to 10 high-quality records from the PhysioNet Sudden Cardiac Death Holter Database (SDDb), we find that $\Delta\tau_s$ and $\Delta\text{excess3}$ are statistically significant in 8/10 cases under phase-shuffle surrogates and, crucially, maintain **sign concordance** even when the direction is opposite to classical variance increase. Records with intermittent pacing (flagged automatically) and atrial fibrillation are included and still exhibit the pattern. The scale ($N=10$ after strict filters) is realistic given that SDDb contains only 23 records total. The relational metrics detect reorganization even when classical variance-based early-warning signals are weak or reversed. Intermittent pacing and atrial fibrillation cases were retained after explicit quality flagging and did not abolish the signal.

These results support a view of pre-VF transitions as context-dependent reorganizations of the relational structure of heart-rate dynamics rather than uniform loss of stability. The framework is fully reproducible, falsifiable, and immediately extensible to other databases.

1. Introduction

- Sudden cardiac death burden.
 - Limitations of univariate EWS (variance, AR(1), DFA, etc.) in cardiac literature.
 - The need for relational / multivariate / ordinal approaches.
 - Systemic Tau (τ_s) as a normalized, windowed measure of coupling change between RR and $|\Delta RR|$.
 - RECD ordinal levels (Bandt-Pompe + conjunctions) and the excess3 statistic as a direct probe of “Level 3” structure.
 - Contribution of this work: first application of τ_s + weighted RECD to spontaneous human VF; explicit handling of real-data pathologies (pacing, AF, annotation quality); strict quality filters on the only publicly available long-term pre-VF Holter collection.
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2. Data and Methods

2.1 SDDDB and record selection

- PhysioNet Sudden Cardiac Death Holter Database (SDDDB): exactly 23 complete Holter recordings (records 30–52), 250 Hz, 2-lead ECG. 18 underlying sinus (4 intermittent pacing), 1 continuously paced, 4 atrial fibrillation. Most have documented VF onset (vfon) comment in .hea; .atr (audited) preferred, .ari (unaudited) fallback.
- Strict inclusion (duration ≥ 12 –15 h, pre-event ≥ 4 –6 h usable after cleaning, clear vfon, low invalid RR/interp_frac $< 6\%$, pacing auto-flagged via comments + KNOWN_PACING + cv <0.06 heuristic).
- Realistic ceiling after filters: ~ 8 –12 high-quality records (SDDDB total only 23). Final N=10 processed: 30,31,32,35,36,38,45,47,50,51 (intermittent pacing in 32/51 correctly detected and retained).

2.2 RR extraction and cleaning

- wfdb.rdann (.atr preferred, .ari fallback).
- RR (ms) = diff(samples)/fs * 1000.
- $250 < \text{RR} < 2000$ ms retained; linear interpolation of outliers.
- Quality metadata exported per record: n_beats, interp_frac, cv_rr, pacing_detected, known_pacing_type.

2.3 Bivariate proxy and Systemic Tau

- $X(t) = [z(\text{RR}), z(|\Delta \text{RR}|)]$
- compute_taus (W_TAU=101, stride=5) $\rightarrow \tau_s$ series.
- Basal vs approach regime comparison (record-specific 3-hour windows anchored to vfon or end-of-recording).
- Phase-shuffle surrogates (n=8) preserving individual spectra but destroying cross-dependence.

2.4 RECD Ordinal Levels + Weighted

- m=3, delay=1, w_phi=101, stride=5 (Bandt-Pompe on bivariate proxy X).
- Φ_1 (coincidence), Φ_2 (persistent relations), Φ_3 + excess3 (proxy of irreducible synergy) with theta3=0.08 (light re-cal for RR noise; original 0.10).
- excess3 raw (continuous) used for means/deltas; high_level3_rate = fraction(excess3 > high_thresh=0.65) (re-cal from synthetic 1.75; remains near-zero here as excess3 ~ 0.30 –0.43).
- Weighted: compute_recd_from_conjunctions with alpha_mode=lambda (relative $|\tau_s|/\text{max}$ per record for small observed deltas; or lowered theta_chaos ~ 0.09).
- Metrics: mean_excess3 (primary), high_level3_rate, frac_contrib3, contribs.
- Same basal/approach windows (record-specific ~ 3 h preserving legacy for 30/35). Welch t-tests; phase-shuffle surrogates on τ_s .

2.5 Quality & reproducibility

- All code in repo; exact command lines recorded.
- Pacing handled explicitly (never silently dropped).

3. Results

3.1 Batch overview (N=10)

record	$\Delta\tau_s$	p_surr	$\Delta\text{excess3}$	p_excess3	interp%	pacing
30	-0.0274	0.00	-0.0118	7.7e-78	0.39	none
31	-0.0001	0.25	-0.0221	1.1e-222	0.43	none
32	-0.0239	0.00	-0.0088	5.1e-108	5.22	intermittent
35	+0.0158	0.00	+0.0036	3.2e-29	0.99	none
36	+0.0786	0.00	+0.0080	1.0e-17	0.09	none (AF)
38	+0.0876	0.00	+0.0414	0.0	6.13	none
45	+0.0066	0.25	+0.0043	0.003	0.39	none
47	-0.0026	0.25	+0.0300	4.3e-68	0.18	none
50	+0.0722	0.00	-0.0033	0.0009	1.07	none (AF)
51	-0.0569	0.00	-0.0322	0.0	0.04	intermittent

Two records with small effect sizes (47 and 50) showed discordant direction between $\Delta\tau_s$ and $\Delta\text{excess3}$; these cases had the smallest absolute deltas in the cohort and are interpreted as borderline transitions where the relational reorganization is subtle.

Key observations - 8/10 records show $|\Delta\tau_s|$ with surrogate $p \leq 0.25$ (6 with $p=0.00$). Sign concordance $\Delta\tau_s$ vs $\Delta\text{excess3}$ in 8/10 (discordant on small deltas in 47 and 50; direction context-dependent: positive for some AF/terminal, negative for paced/intermediate). - Strongest: 38 (+0.0876 τ / +0.0414 ex3), 36 (+0.0786), 51 (-0.0569 / -0.0322), 50 (+0.0722 τ). - Paced (32,51) and AF (35,36,50) retained and produce strong signals when flagged. - interp median ~0.4%; max 6.1% (38) still usable. - high_level3_rate (thresh=0.65 post re-cal) remains 0.0 (excess3 values ~0.30 basal to ~0.43 approach); continuous excess3 is the robust, sensitive metric. Re-cal primarily future-proofs the pipeline.

3.1b Light re-calibration (theta3 / high / λ)

Observed on the 10 RR series: $|\Delta\tau_s|$ median $\approx$ 0.026 max $\approx$ 0.088; basal excess3 $\approx$ 0.30–0.35. - theta3: 0.10 $\rightarrow$ **0.08** (slightly more sensitive for noisy physiology). - high_thresh: 1.75 $\rightarrow$ **0.65** (makes rate metric defined; still yields ~0 here). - The binary high_level3_rate remained zero at the recalibrated threshold (0.65) because observed excess3 values in these noisy RR series ranged between ~0.30 (basal) and ~0.43 (approach). Consequently, the continuous mean_excess3 and its delta are used as the primary metrics; the rate metric is retained for future use on cleaner or longer recordings. - λ : switched to record-relative scaling $|\tau_s|/\max|\tau_s|$ (or lowered absolute ~0.09) to activate α variation given small observed $|\tau_s| \ll$ synthetic 0.41. Deltas for mean_excess3 nearly unchanged (raw excess3 independent of theta3); weighted nearly identical to unweighted when both exist. Sensitivity table in supp. The re-cal improves interpretability without altering main sign-concordance conclusions.

3.2 Classic EWS vs relational

- Variance almost always increases ($\uparrow$).
- AR(1) almost always decreases ($\downarrow$) — opposite to naive CSD.
- τ_s and excess3 capture the *direction of the relational reorganization* (terminal vs intermediate vs AF-triggered).

3.3 Weighted RECD

- On current data $|\tau_{sl}| \ll$ synthetic threshold (median $|\Delta\tau_{sl}| \approx 0.026$, max 0.088) $\rightarrow \lambda$ nearly constant across time.
- Therefore weighted and unweighted excess3 deltas are numerically almost identical when both computed (e.g. record 38: unweighted $\Delta = 0.04151$, weighted $\Delta = 0.04136$).
- Weighted RECD (full λ from $|\tau_{sl}|$) successfully completed on 6/10 records (30,31,32,35,38,51). Records 36,45,47,50 use unweighted excess3 from the RECD levels step. A `has_weighted` flag is carried in the consolidated table. When both versions exist, deltas are nearly identical (e.g. record 38).
- `frac_contrib3` remains high and relatively stable (Nivel 3 already dominant). The lambda-driven α ramp will become more relevant after the proposed re-calibration of the `lrl` activation threshold.

3.4 Batch figures (see figures/batch/)

- $\Delta\tau_s$ and $\Delta\text{excess3}$ bar comparison
 - $-\log_{10}(p)$ heatmap
 - Quality (interp % + “P” flags for paced)
-

4. Discussion

- Pre-VF dynamics are **context-dependent** (sign of τ_s and excess3 flips).
 - The relational/ordinal view detects reorganization even when univariate variance/AR(1) behave “classically”.
 - Pacing and AF do not destroy the signal when properly flagged.
 - Limitations of SDDb (N=23 total, heterogeneous annotation quality, sparse clinical metadata) are real but do not invalidate the scale for a first preprint in complex-systems / network-physiology venues.
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5. Limitations

- SDDb contains only 23 records total (physionet.org/content/sddb/1.0.0/). After strict quality/duration/pre-event/vfon filters, realistic max high-quality $N \approx 8-12$ (here $N=10$). 18 sinus (4 intermittent pacing), 1 continuous paced, 4 AF; vfon markers present in VF cases via .hea comments.
 - Record 32 (intermittent pacing, highest interpolation fraction at 5.22%) was retained after automated quality flagging but exhibited noisier dynamics in the approach window; its inclusion did not reverse the overall sign-concordance pattern.
 - Annotation: .atr (preferred) audited in subset; .ari unaudited fallback for some (e.g. 38,44+). Pacing detected robustly (comment + known + cv heuristic) but RR interpretation in paced segments requires caution.
 - Clinical metadata extremely sparse (age/gender/history/meds often unknown or partial; no drug dosages/timing).
 - Single public database with retrospective collection (1980s Boston); heterogeneous substrates. External validation on independent VF Holter collections mandatory before any clinical translation.
 - $|\tau_{sl}|$ and excess3 magnitudes small (typical physiology); re-cal of synthetic-derived thresholds was required (and implemented). `high_level3_rate` at 0.65 still uninformative on these data.
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6. Conclusions & Outlook

Systemic Tau combined with ordinal RECD provides a reproducible, sign-concordant early-warning signature of spontaneous VF that is invisible or reversed by classical univariate EWS. These findings provide real-world physiological support for the Systemic Tau and RECD framework as tools capable of detecting context-dependent relational reorganizations that precede critical transitions, even in noisy biological signals where classical univariate early-warning signals fail or reverse. The next steps are (i) threshold re-calibration tuned to real RR statistics, (ii) expansion to additional records/databases, and (iii) preprint submission.

Appendix / Commands (reproducibility)

1. Download / extract (if needed)

```
python3 code/download_sddb_records.py --records 30,32,35,36,38,44,45,47,50,51,52
```

```
python3 code/extract_rr.py --record 38 # etc; produces rr*_clean.npz + quality keys
```

2. Full batch (original params) or with light re-cal

```
python3 code/run_cctp_batch.py --records 30,31,32,35,36,38,45,47,50,51
```

Re-cal (theta=0.08, high=0.65, relative lambda):

```
python3 code/run_cctp_batch.py --records 30,32,35,36,38,50,51 --theta3 0.08 --high-thresh 0.65 --la
```

3. Individual re-cal (e.g. strongest)

```
python3 code/run_recd_on_rr.py --record 38 --theta3 0.08 --high-thresh 0.65
```

```
python3 code/run_recd_weighted_on_rr.py --record 38 --theta3 0.08 --high-thresh 0.65 --lambda-relat
```

4. Regenerate summary + batch figs + inspect

```
python3 code/run_cctp_batch.py --records 30,31,32,35,36,38,45,47,50,51
```

```
python3 -c "
```

```
import pandas as pd
```

```
df = pd.read_csv('results/cctp_batch_summary.csv')
```

```
print(df[['record','delta_tau','delta_excess3','delta_high_level3','interp_frac','pacing_detected',
```

```
print('Sign concordance check...')
```

```
"]
```

Supplementary figures list (preprint/figures/ or figures/)

- Batch: batch_delta_tau_excess3.png, batch_significance.png, batch_quality_interp.png

- Per-record key (38, 50, 36, 51, 30, 35): 01_rr_full.png, 05_rr_full_with_approach.png, 06_ews_panels.png, 10_irregularity_and_tau.png, 16_recd_excess3.png, 17_recd_excess3_box.png, 18/19 weighted (when available).

- Sensitivity: original vs re-cal (theta 0.10/1.75 vs 0.08/0.65 + rel λ) on mean_excess3 deltas (nearly identical).

Data & code availability All scripts, records_inventory.csv, selected_records.txt, and exact commands are in the repository. SDDB public via PhysioNet (DOI 10.13026/C2W306). Cite Greenwald thesis + PhysioNet.

Status of this draft: Preprint-ready v0.9 (N=10 authoritative table + batch figs + full Methods + re-cal + SDDB Limitations + repro commands + final polish: refined title, abstract pacing/AF sentence, borderline 47/50 note, Record 32 limitation). Ready for medRxiv / Chaos / Frontiers in Network Physiology.